

Practice Quiz 04C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 04

Set identities by logical laws; unions of families and power sets.

Name a definition or law at every step of part (a).

1. (a) Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ by a chain of equivalences beginning with membership in the left side.

Solution For an arbitrary object x , the definitions of difference and union give

$$x \in A \setminus (B \cup C) \iff x \in A \wedge \neg(x \in B \vee x \in C).$$

By De Morgan's law this is equivalent to

$$x \in A \wedge (x \notin B \wedge x \notin C).$$

The idempotent and associative laws make the last statement equivalent to

$$(x \in A \wedge x \notin B) \wedge (x \in A \wedge x \notin C).$$

By the definitions of difference and intersection, this is equivalent to $x \in (A \setminus B) \cap (A \setminus C)$. Therefore the sets are equal. ■

- (b) Let $\mathcal{F} = \{\{1, 2\}, \{2\}\}$. Compute $\bigcup \mathcal{F}$ and then $\mathcal{P}(\bigcup \mathcal{F})$.

Solution $\bigcup \mathcal{F} = \{1, 2\}$, so

$$\mathcal{P}(\bigcup \mathcal{F}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}. \quad \blacksquare$$